Department of Mathematics, IIT Madras

MA 2102 - Differential Equations, First Semester 2026-27

Assignment-3 : Cauchy-Euler Equations and Higher Order Differential Equations

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1. Find the general solution of the following differential equations.
   (a) $x^2y'' + 3xy' + 10y = 0$; (b) $2x^2y'' + 10xy' + 8y = 0$;
   (c) $x^2y'' + 2xy' - 12y = 0$; (d) $4x^2y'' - 3y = 0$;
   (e) $x^2y'' + 2xy' - 6y = 0$; (f) $x^2y'' + 2xy' - 6y = 0$;
   (g) $x^2y'' + xy' - 2y = 0$; (h) $x^2y'' + xy' - 16y = 0$.

2. Solve the following IVPs.
   (a) $x^2y'' + xy' + y = 0; \quad y(1) = 1,\ y'(1) = 1.$
   (b)$4x^2y'' + 12xy' + 3y = 0; \quad y(1) = 1,\ y'(1) = \dfrac{3}{2}.$

3. Solve: *(Hint: Use method of undetermined coefficients or variation of parameters for the particular integral after the appropriate substitution such as $x = e^t$.)*
   (a) $x^2y'' - 3xy' + 4y = 2x^2$; (b) $x^2y'' + 2xy' - y = 3x^3\cos(\log x)$;
   (c) $y'' + \dfrac{5}{x}y' + \dfrac{3}{x^2}y = e^{-x}$; (d) $x^2y'' + 3xy' + y = \dfrac{1}{(1-x)^2}$;
   (e) $y'' + \dfrac{1}{x}y' = \dfrac{12\log x}{x^2}$; (f) $(2+x)^2y'' + (2+x)y' + 4y = 2\sin(2\log(2+x))$;
   (g) $(5+2x)^2y'' + 6(5+2x)y' + 8y = 8(5+2x)^2$; (h) $(x+a)^2y'' - 4(x+a)y' + 6y = x$.

4. Show that the change of variable $t = ax+b$ transforms the equation$b_0(ax+b)^2y''+b_1(ax+b)y'+b_2y = 0$ into a Cauchy-Euler equation. Use this result to find the general solution of the differential equation $(x-3)^2y'' + 3(x-3)y' + y = 0, \quad x > 3$. **Ans.** $y = [c_1 + c_2\ln(x-3)](x-3)^{-1}$.

5. Find a second order homogeneous linear ODE for which the given functions are solutions. Show linear independence by the Wronskian. Solve the corresponding IVPs.
   (a) $x^2,\ x^2\ln x; \quad y(1) = 4,\ y'(1) = 6.$ **Ans.** $x^2y'' - 3xy' + 4y = 0$
   (b) $x^2\cos(\ln x),\ x^2\sin(\ln x); \quad y(1) = 3,\ y'(1) = 8$
   (c) $\dfrac{1}{x},\ x^3; \quad y(2) = -1,\ y'(2) = 1.$

6. Consider the differential equation $y'' + 2xy' + (\exp(x)\sin x)y = 0$. Let y_1 and y_2 be solutions of the differential equation satisfying $y_1(0) = 2,\ y_2(0) = -3,\ y_1'(0) = 1,\ y_2'(0) = 2$. Then,(i) find the Wronskian of y_1 and y_2 (ii) discuss the linear independence of y_1 and y_2.

7. Without computing the two linearly independent solutions y_1 and y_2, compute their Wronskian for the differential equation $x^2y'' - x(x+2)y' + (x+2)y = 0, \qquad x > 0.$

8. Use the method of variation of parameters to find the general solution:
   (a) $x^2y'' + xy' - y = x^2\exp(-x)$ **Ans.** $y(x) = c_1x + c_2x^{-1} + \exp(-x)(1 + x^{-1})$.
   (b)$(x-3)^2y'' + (x-3)y' + y = x$ **Ans.** $y(x) = c_1\cos[\ln(x-3)] + c_2\sin[\ln(x-3)] + x^2 + \dfrac{3}{2}$.

9. Find the general solution of the following differential equations. *(Use the method of undetermined coefficients for particular solutions.)*
   (a) $y''' - 3y'' + 4y' - 2y = x^3 - 2x$; (b) $y''' + y = 4e^{-x}$;
   (c) $y^{(4)} + 4y''' + 6y'' + 4y' + y = x^2e^{-x}$; (d) $y^{(4)} - y = 8\sin x$;
   (e) $y^{(4)} + 5y'' + 4y = \cos 3x$; (f) $y^{(4)} + 2a^2y'' + a^4y = x\sin ax$;
   (g) $y^{(4)} + 2y''' + 2y'' + 2y' + y = xe^{-x}$; (h) $y^{(4)} + 2y''' - 2y'' - 6y' + 5y = x\cos x$;
   (i) $y^{(4)} + y''' - 3y'' - 5y' - 2y = 2e^{2x}$; (j) $y''' + 2y'' - y' = 3x^2 - 2x + 1$;
   (k) $y''' - 3y'' + 2y' = 10 + 42e^{3x}$; (l) $y''' - 6y'' + 11y' - 6y = 3e^{2x}$.

10. Solve the following third order Cauchy-Euler equations:
   (a) $x^3y''' + 3x^2y'' = 0$; (b) $x^3y''' + 2x^2y'' + xy' - y = 0$;
   (c) $x^3y''' + x^2y'' - 2xy' + 2y = 0$; (d) $x^2y''' + 3xy'' - 2y' = 0$.

11. Find the general solution of the following:
   (a) $x^3y''' + x^2y'' - 2xy' + 2y = x^{-2}$;
   (b) $4x^3y''' + 3xy' - 3y = 10$;
   (c)$x^3y''' + x^2y'' - 2xy' + 2y = x\ln x$;
   (d) $y''' - 3y'' + 3y' - y = x^{1/2}e^x$.

12. Solve the following IVPs:
   (a) $y^{(4)} - 5y'' + 4y = 10e^{-3x}$; $y(0) = 1,\ y'(0) = y''(0) = y'''(0) = 0$.
   (b) $y^{(4)} + 5y'' + 4y = 90\sin 4x$; $y(0) = 1,\ y'(0) = 2,\ y''(0) = -1,\ y'''(0) = -32$.
   (c) $x^3y''' + xy' - y = x^2$; $y(1) = 1,\ y'(1) = 3,\ y''(1) = 14$.
   (d) $y''' - 4y' = 10\cos x + 5\sin x$; $y(0) = 3,\ y'(0) = -2,\ y''(0) = -1$.
   (e) $y''' + 3y'' + 3y' + y = 8\sin x$; $y(0) = -1,\ y'(0) = -3,\ y''(0) = 5$.

13. In determining the drag on a small sphere moving at a constant speed through a viscous fluid, it is necessary to solve the differential equation

$$x^3y^{(4)} + 8x^2y''' + 8xy'' - 8y' = 0.$$

   Thus, compute the drag of this small sphere.